The Effect of Background Noise on Cognitive Performance (Reaction Time)

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Software Used: PsychoPy (v2026.1.3)

Abstract

This study investigated the effect of background noise (simulated café environment) on visual reaction time performance. Contrary to the initial hypothesis that noise would act as a distraction and slow reaction times, the results indicated a trend where background noise was associated with **faster reaction times**.

These findings suggest that moderate ambient noise may enhance alertness (arousal) during simple visual tasks.

1. Introduction

Background

In modern work environments, individuals are frequently exposed to ambient background noise. Previous research suggests that noise can act as a stressor and impair cognitive performance.

However, the **Yerkes–Dodson Law** proposes that moderate levels of arousal (such as background noise) may actually enhance performance, particularly for simple tasks.

Hypothesis

It was hypothesized that participants would exhibit **slower reaction times** in the *Noise* condition compared to the *Silence* condition due to auditory distraction.

2. Methodology

2.1 Participants

Data was collected from **3 participants** using convenience sampling:

- test1
- test2
- test3

2.2 Experimental Design

- **Independent Variable (IV):** Audio Condition

    - Silence
    - Noise

- **Dependent Variable (DV):**

    - Reaction Time (seconds)

- **Design Type:**

    - Within-subjects (Repeated Measures)

2.3 Apparatus & Materials

- **Software:** PsychoPy (v2026.1.3)
- **Stimulus:** Green circle following fixation cross (+)
- **Audio:**

    - Noise condition: cafe_noise.wav (10 seconds)
    - Silence condition: No audio

2.4 Procedure

1. Participants read instructions and pressed **Space** to begin.
2. A fixation cross appeared for **0.5 seconds**.
3. A green circle was displayed.
4. Participants pressed **Spacebar** as quickly as possible.
5. A total of **20 randomized trials** were conducted (Silence + Noise).

3. Results

Data was preprocessed to remove:

- Missed trials
- Anticipation responses *(RT < 0.1s)*

3.1 Descriptive Statistics

Participant	Condition	Mean Reaction Time (s)	Observation
test1	Silence	0.319 s	Fast performance
test1	Noise	0.237 s	**Faster in noise**
test2	Silence	1.107 s	Slower overall
test2	Noise	1.001 s	**Faster in noise**
test3	Silence	1.338 s	Slowest performance

test3	Noise	0.850 s	**Significantly faster in noise**

3.2 Statistical Analysis

Independent Samples *t-tests* were conducted:

- **Participant test1:**
  $t(18) = 1.92$, **$p = 0.07$** → Not significant (trend observed)
- **Participant test2:**
  $t(14.5) = 0.64$, **$p = 0.53$** → No significant difference
- **Participant test3:**
  $t(15) = 2.40$, **$p = 0.028$ → Statistically significant**

Summary

- All participants showed **faster reaction times under noise**
- One participant demonstrated **statistical significance**
- Overall trend supports improved performance in noise

4. Discussion

4.1 Interpretation

The hypothesis was **rejected**. Contrary to expectations, all participants performed better in the Noise condition.

This supports the **Arousal Theory**, indicating that moderate background noise may:

- Reduce mind-wandering
- Increase alertness
- Improve response speed

4.2 Limitations

- **Small Sample Size:** Only 3 participants
- **Data Quality Issues:** Missed trials and one excluded outlier
- **Technical Constraints:** Audio handled via pygame (lower timing precision than dedicated hardware)

4.3 Conclusion

The findings suggest that background noise does not necessarily impair cognitive performance.

For simple reaction-based tasks, ambient noise may function as a **performance enhancer** rather than a distraction.